Self-Balancing Robot

Based on the ESP32 microcontroller and MPU6050 inertial unit

Course: Modelling and Control of Electric Drives
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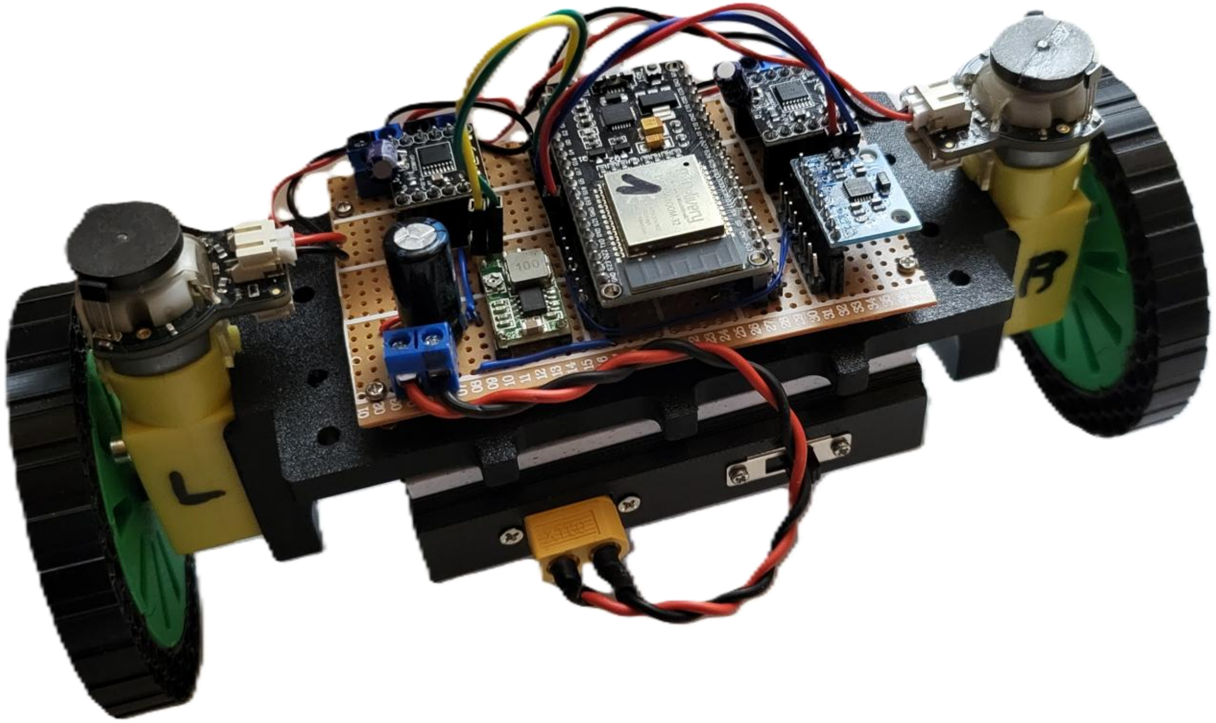
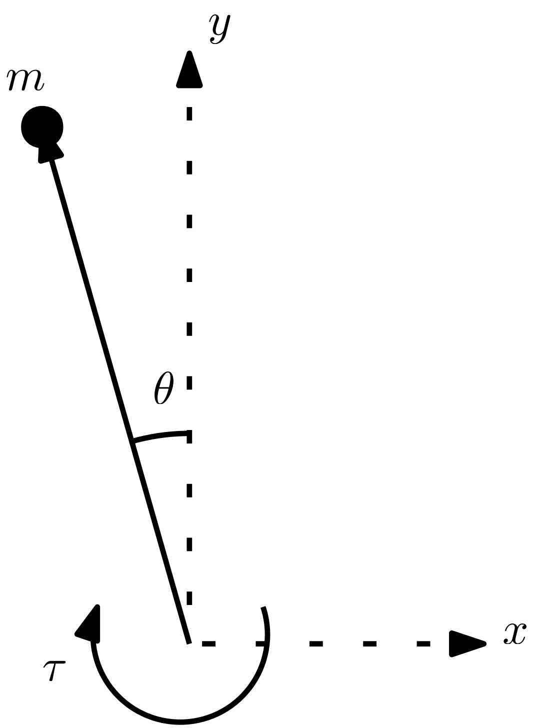
Overview

- System model
  - Motor Inductance Estimation
- Implementation
  - Schematics
  - FSM and Timing
  - Angle Measurement



System Model



Approximate Description of the System

$$ml^2\ddot{\theta}(t) = mgl\theta + \tau(t) \Rightarrow G_{pendulum}(s) = \frac{\theta(s)}{\tau(s)} \approx \frac{1}{ml^2} \cdot \frac{1}{s^2 - \frac{g}{l}}$$

$$\begin{cases} v(t) = R_a i_a(t) + L_a \dfrac{di_a(t)}{dt} + K_e\Phi\, \omega(t) \\ \tau(t) = K_e\Phi\, i_a(t) \end{cases} \Rightarrow G_{motor}(s) = \frac{\tau(s)}{v(s)} = k_G \cdot \frac{K_e\Phi}{R \cdot s_L}$$

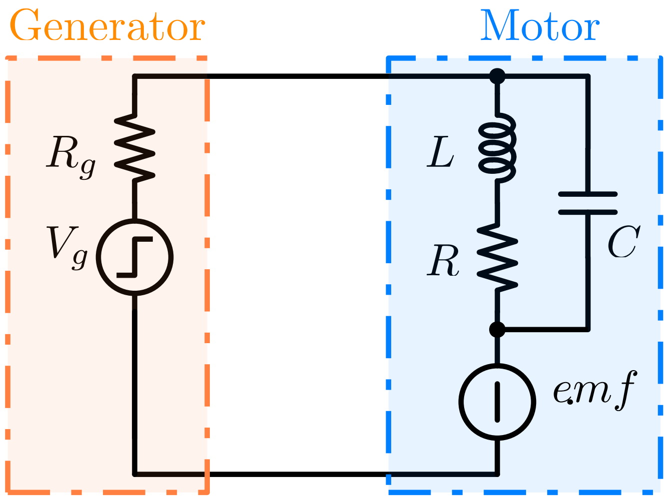
DC Motor Parameters

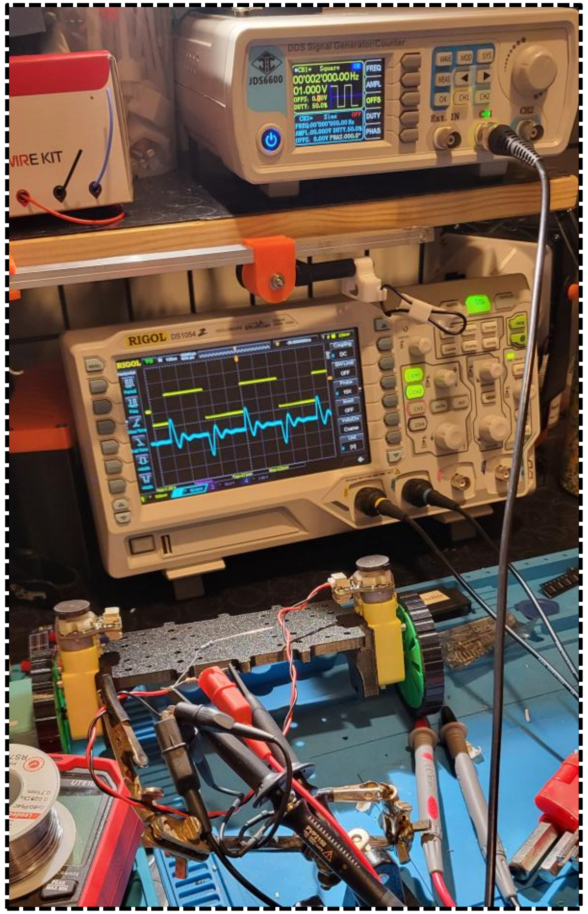
- Nominal voltage $V_N = 6V$
- Armature resistance $R = 3.5\ \Omega$
- ***Armature inductance*** $L = ?$
- *Gear-ratio* $k_G = 120$
- *Torque constant* $K_e\Phi = 0.336$

Estimated Pendulum Parameters

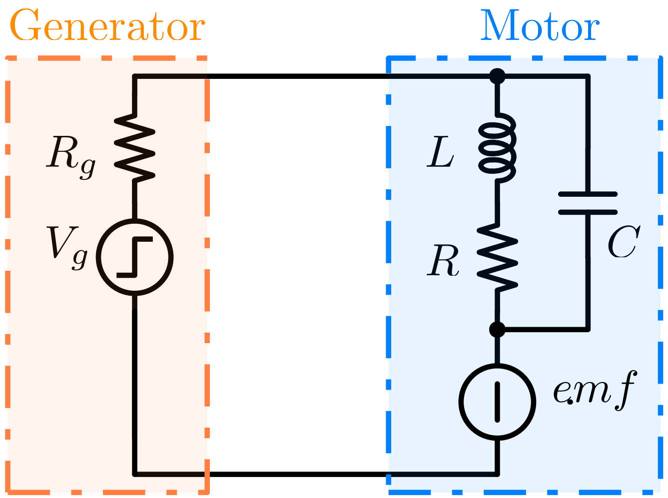
- Mass $m = 0.450\ kg$
- Length $l = 0.08\ m$

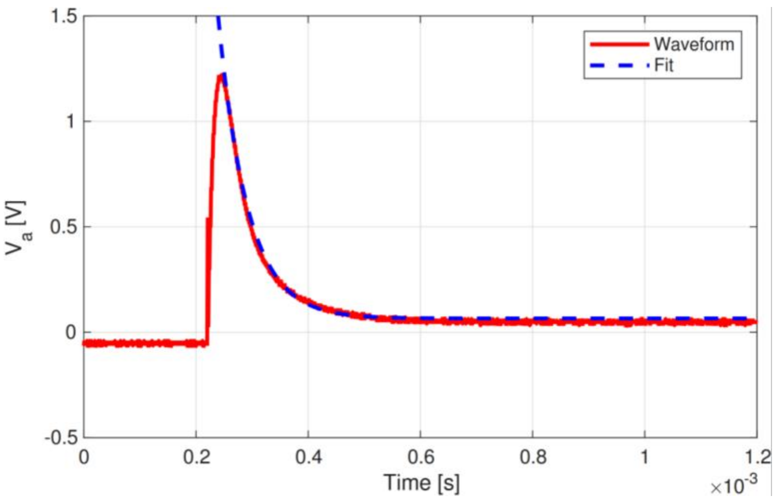
Motor Inductance Estimation





Motor Inductance Estimation



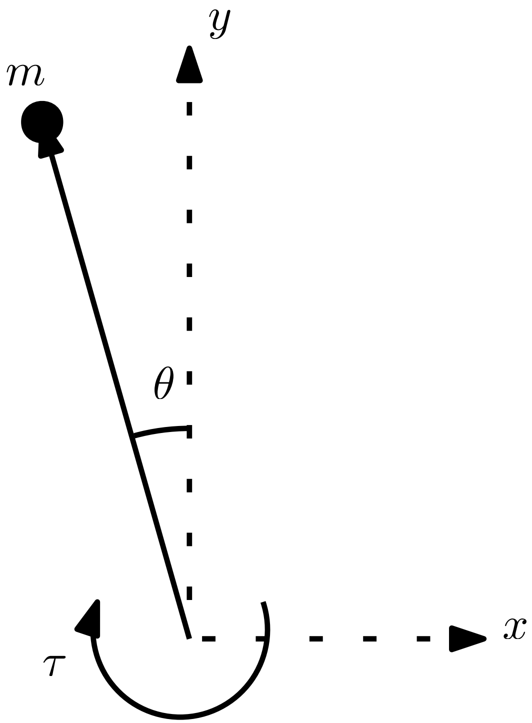


DC Motor Parameters

- Armature resistance $R = 3.5\ \Omega$
- **Armature inductance L $= 5\ \mu H$**
- *Gear-ratio* $k_G = 120$
- *Torque constant* $K_e \Phi = 0.336$

$$v(t) = a \cdot \exp(b \cdot t) + \frac{R}{R + R_g} \Rightarrow b = -\frac{1}{\tau} \Rightarrow L = -\frac{R}{b}$$

System Model



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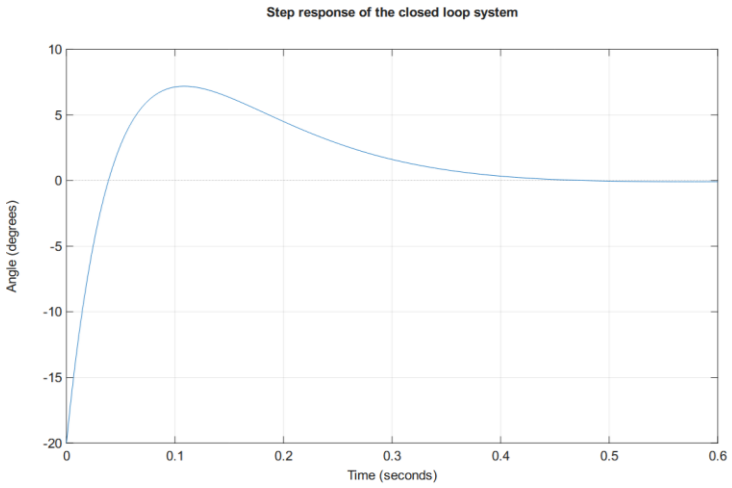
PID Regulator Initial Design

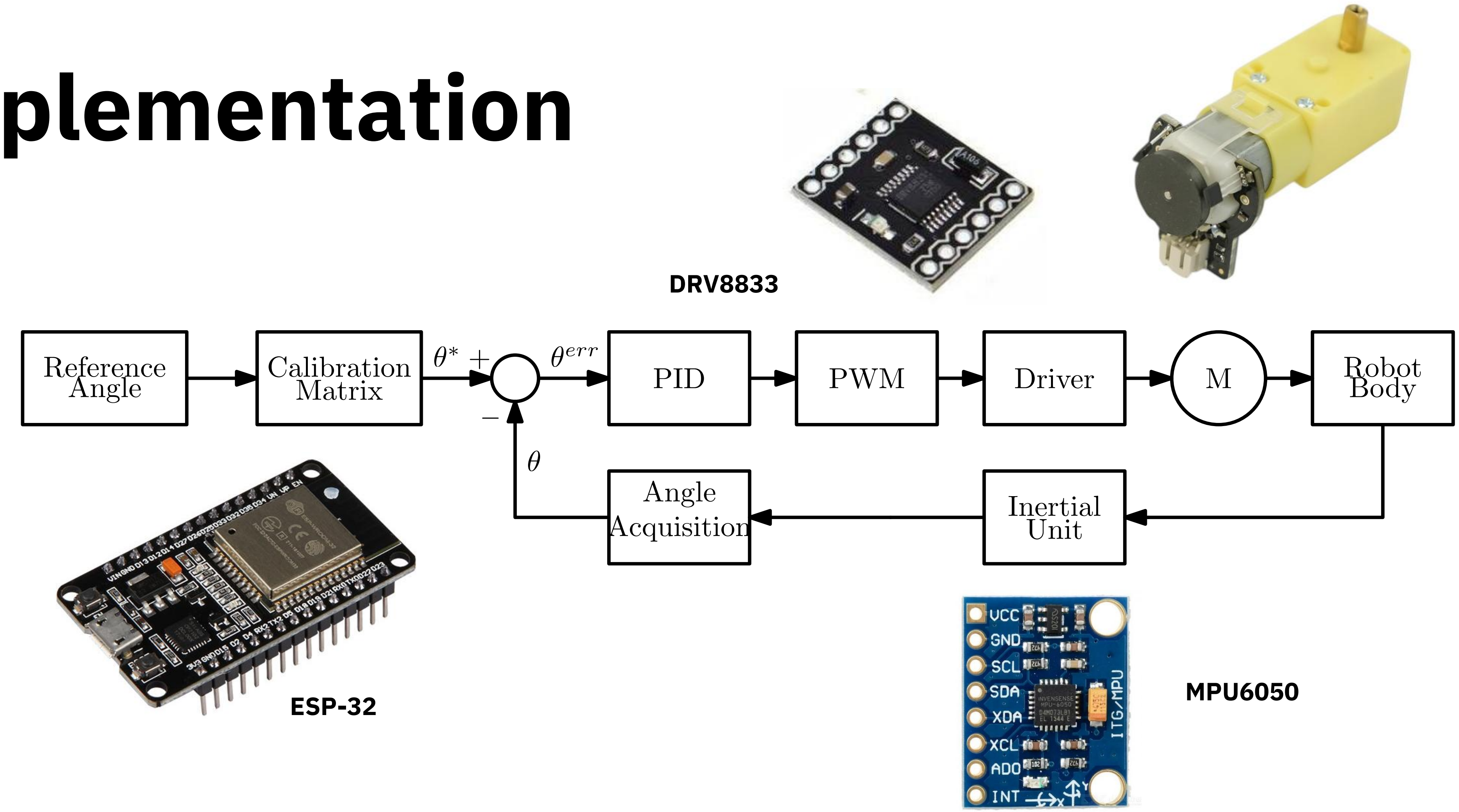
Target rise time: 50 *ms*

$$G_{PID} = K_P + \frac{K_I}{s} + K_D s$$

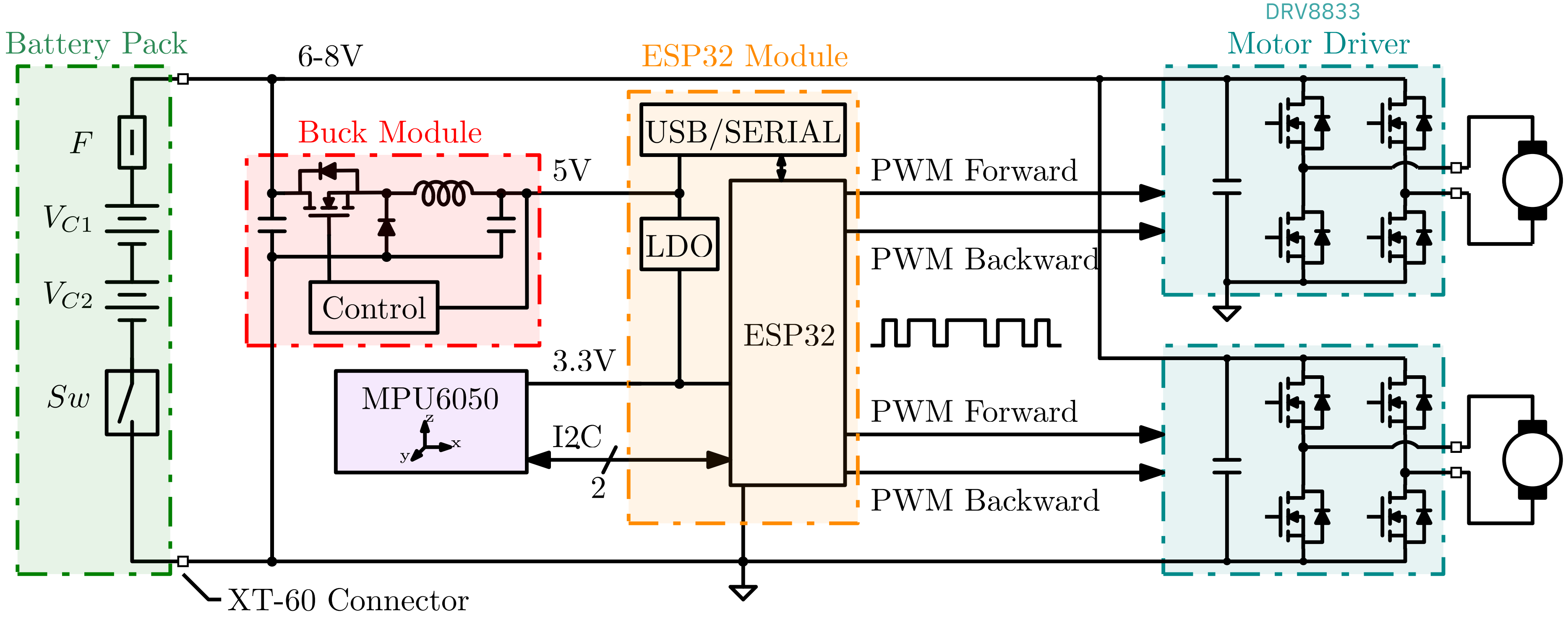
- $K_P = \mathbf{0.0361}$
- $K_I = \mathbf{0.1412}$
- $K_D = \mathbf{0.0023}$



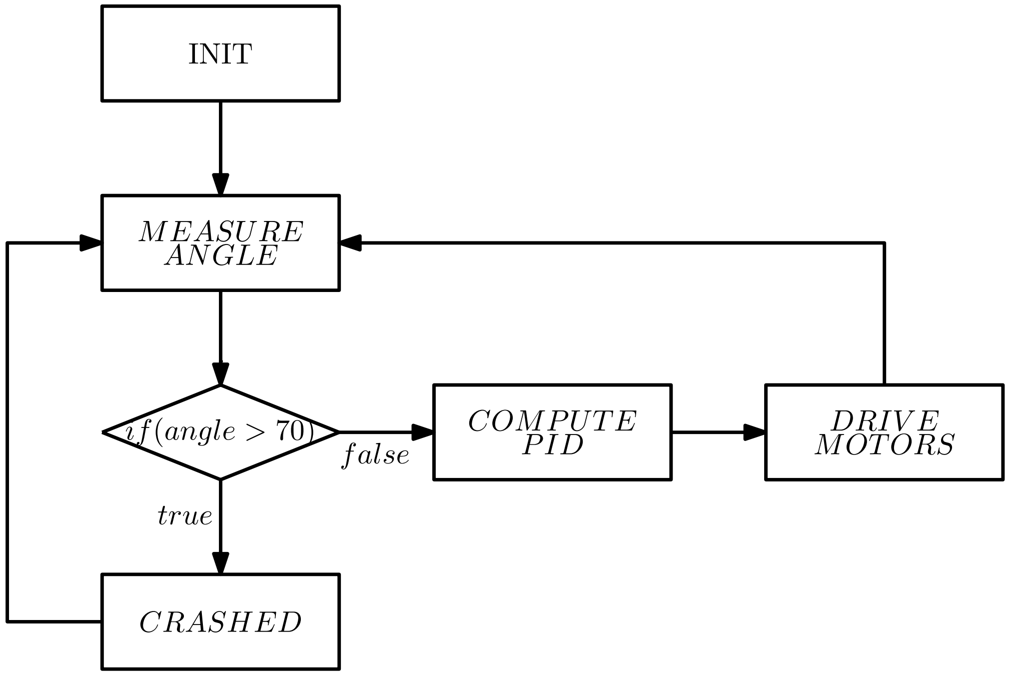
Implementation



Overall Simplified Schematic



Finite State Machine and Timing



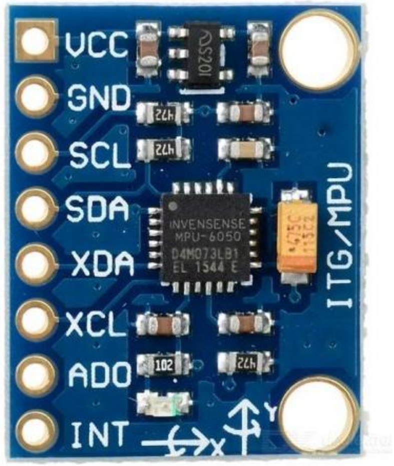
Loop Performance

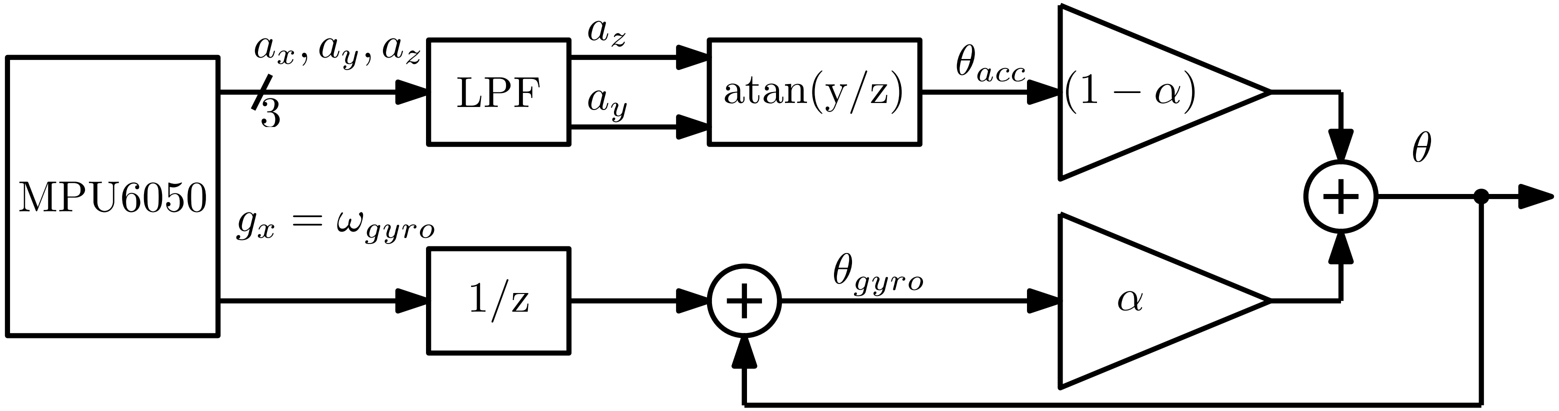
Loop execution period: $4\ ms$ (interrupt based)
Angle sampling rate: $250\ Hz$
Guaranteed rise time: $23\ ms$ (against $50\ ms$ of initial design)
PWM frequency: $100\ Hz$

States:

1. Init
2. Measure Angle
3. Compute PID
4. Drive Motors
5. Crashed

Angle Measurement





Accelerometer: provides acceleration along *x,y,z.* Knowing the initial position, the pitch angle can be determined. **Noisy, requires LPF – too slow**;
Gyroscope: outputs angular velocity. Integration provides the angle relative to the starting point. **Fast but imprecise and tends to drift away**;
Complementary Filter: best of both worlds, gyro for **fast response to transients**, accelerometer for **low frequency and bias:**

$$\theta[k] = \alpha\big(\theta[k-1] + \omega_{gyro}[k]\Delta t_s\big) + (1-\alpha)\theta_{\mathrm{acc}}[k]$$

Manufacturing

Tires Material and Geometry

The objective is achieving higher friction coefficient to avoid tire slip.

1. **TPU** material: **elastic** 3D printing filament, better than the alternatives, but produces slick surfaces – prone to slipping.
2. **Honeycomb** structure: **deforms** under the weight of the robot and provides **additional grip**

Mechanical Assembly Design

- Modeled using SOLIDWORKS CAD
- 3D printed in **PETG** (main body) and **TPU** (tires)

100% infill

15% infill
2mm walls

10% infill
1mm walls

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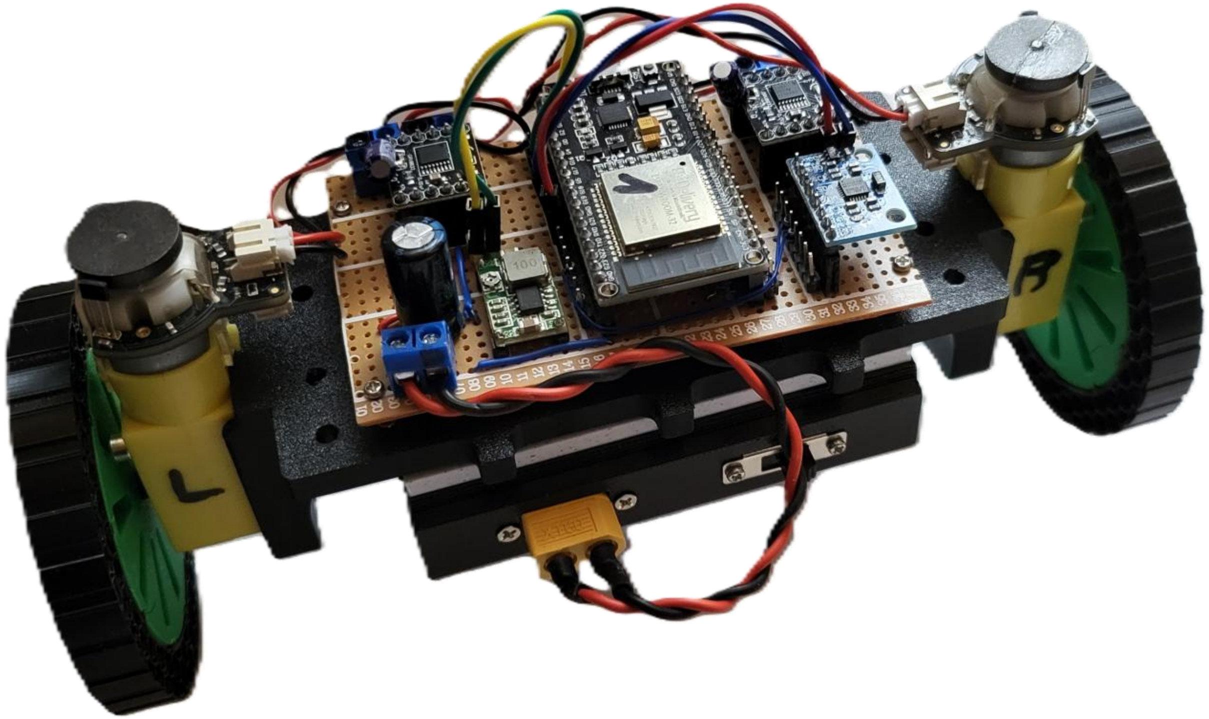
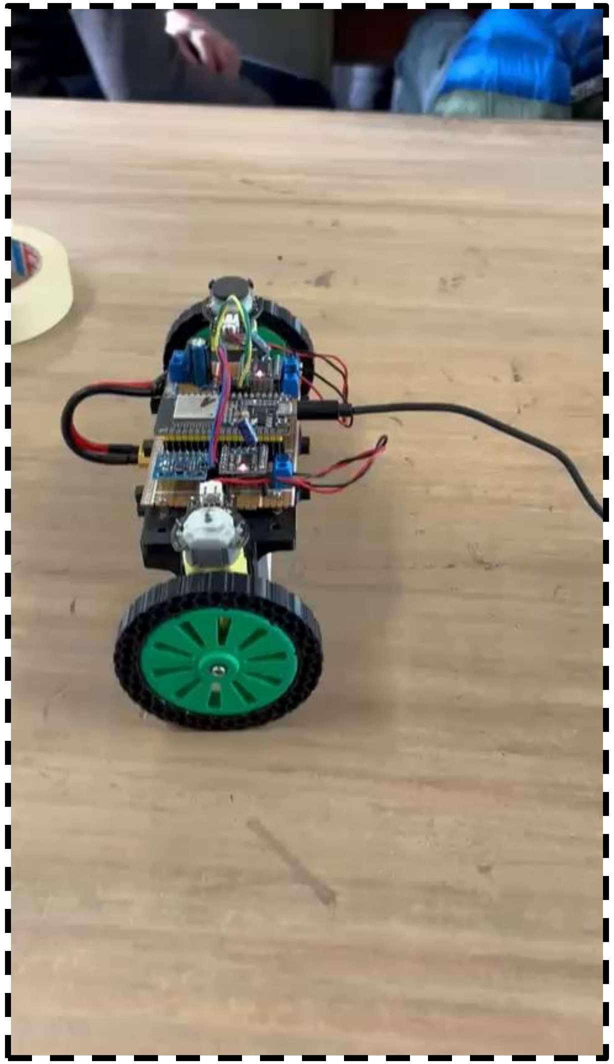
Thank you for listening





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